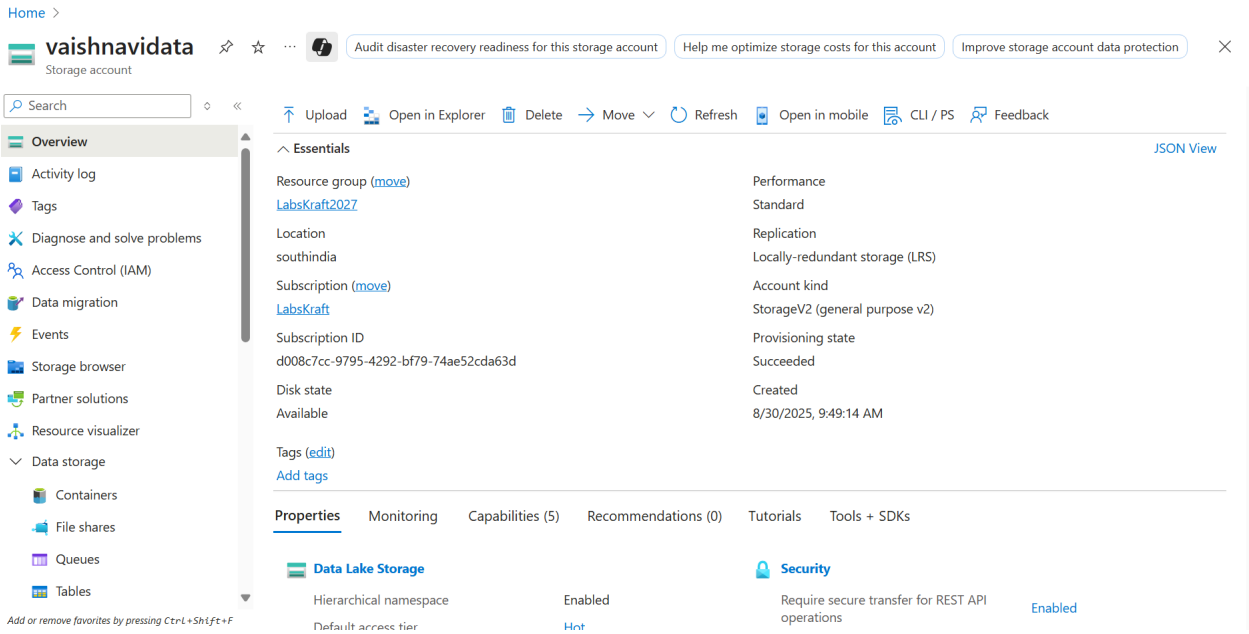
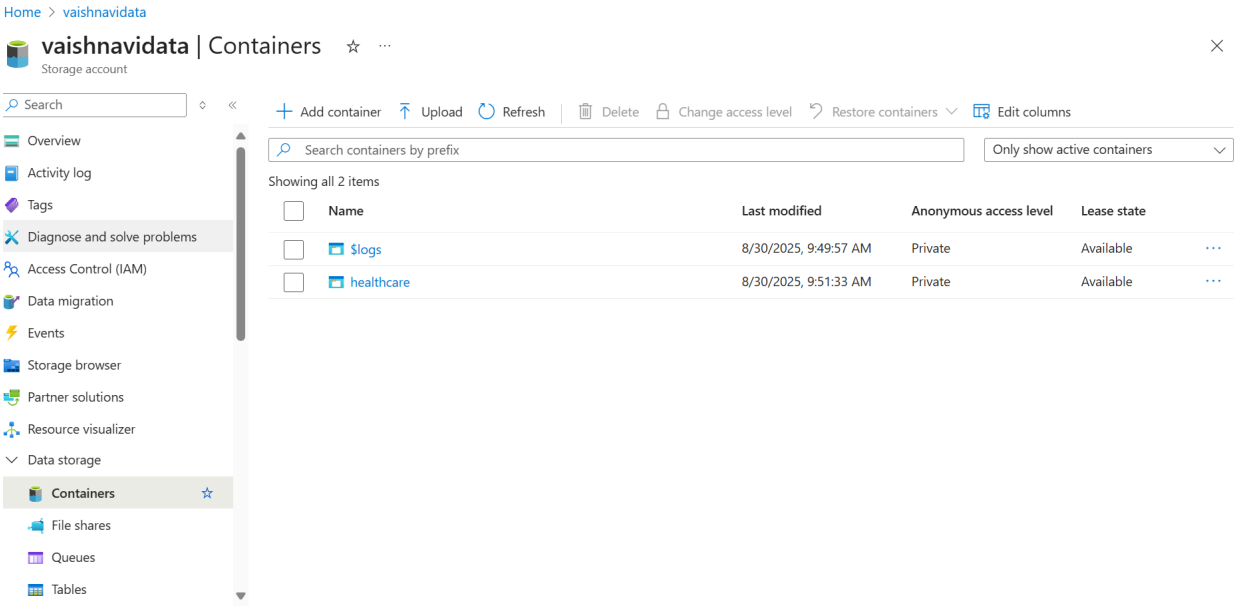
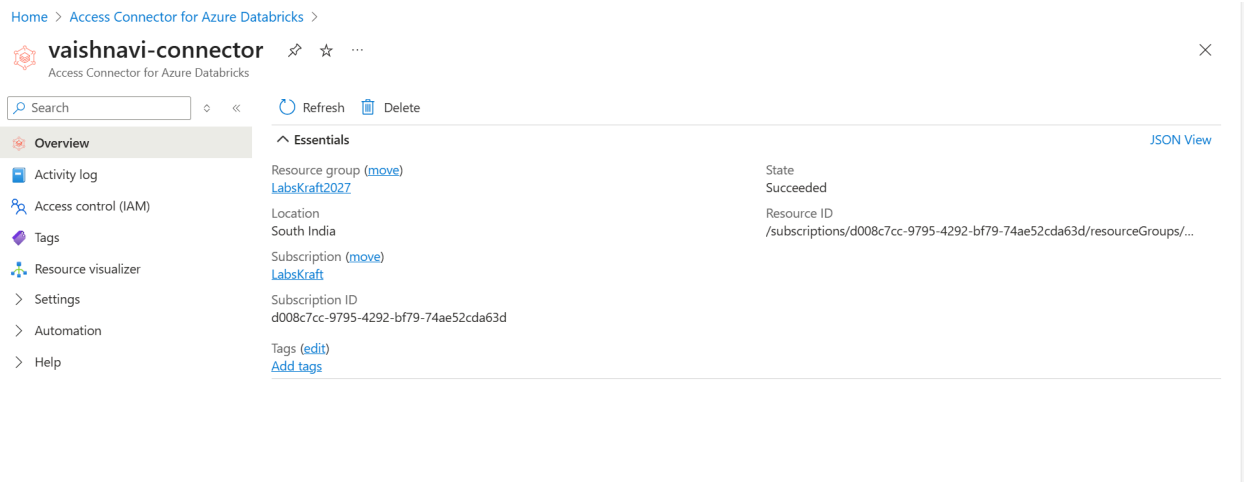
Step -1 Created a Storage Account with hierarchical namespace enabled.



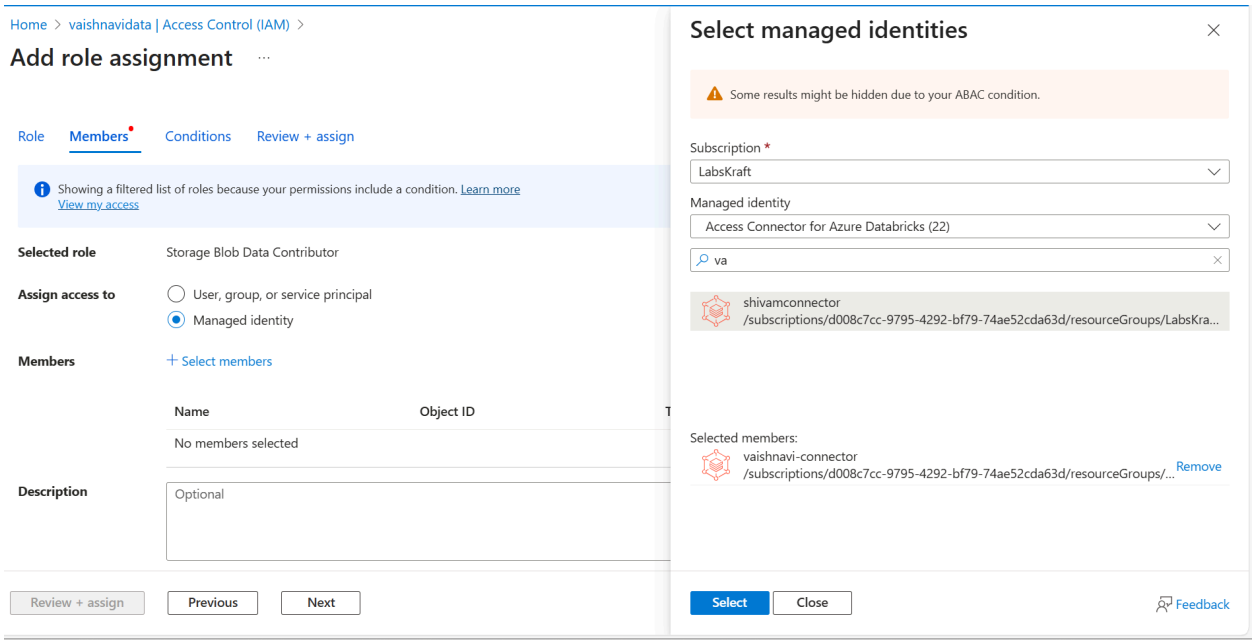
Step - 2 Created a container named healthcare and uploaded the files in the respective folders.



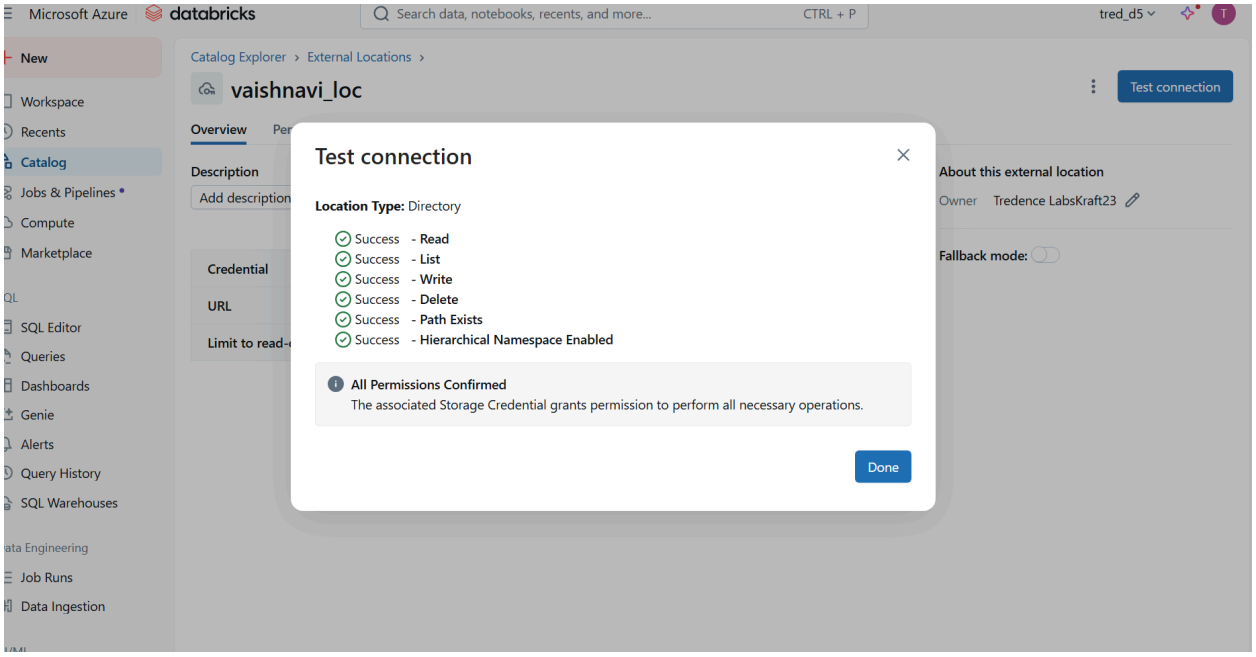
Step -3 Create Access Connector and copy the resource id to add in storage credentials while creating external location for the unity catalog.



Step-4 Give IAM permission of Storage Blob Data Contributor to the managed identity->Access Connector->vaishnavi-connector.



Step - 5 Create external location and unity catalog -> healthcare_vaishnavi



Step -6 Creating bronze, silver and gold tables.

```
%sql
CREATE SCHEMA IF NOT EXISTS healthcare_vaishnavi.bronze;
CREATE SCHEMA IF NOT EXISTS healthcare_vaishnavi.silver;
CREATE SCHEMA IF NOT EXISTS healthcare_vaishnavi.gold;

import json
from pyspark.sql.types import *
from pyspark.sql.functions import *

# ----------------------------------
# Step 0: Setup
# ----------------------------------
catalog = "healthcare_vaishnavi"
schema  = "bronze"

spark.sql(f"USE CATALOG {catalog}")
spark.sql(f"CREATE SCHEMA IF NOT EXISTS {schema}")
spark.sql(f"USE {schema}")

# Raw data paths
```

```
raw_labs_data =
"abfss://healthcare@vaishnavidata.dfs.core.windows.net/labs_data"
raw_patients_data =
"abfss://healthcare@vaishnavidata.dfs.core.windows.net/patients_data"
raw_visits_data =
"abfss://healthcare@vaishnavidata.dfs.core.windows.net/visits_data"
raw_vitals_data =
"abfss://healthcare@vaishnavidata.dfs.core.windows.net/vitals_data"
raw_patient_doctor =
"abfss://healthcare@vaishnavidata.dfs.core.windows.net/patient_doctor"
```

Bronze Table

1) **Bronze Labs Data**

```
bronze_labsdata = (spark.readStream
    .format("cloudFiles")
    .option("cloudFiles.format", "csv")
    .option("header","true")
    .option("cloudFiles.schemaLocation",
"abfss://healthcare@vaishnavidata.dfs.core.windows.net/schema/bronze_labs
data")
    .option("cloudFiles.inferColumnTypes", "true")
    .option("cloudFiles.schemaHints","date DATE")
    .load(raw_labs_data)
    .withColumn("ingest_file", col("_metadata.file_path"))
)

(bronze_labsdata.writeStream
   .format("delta")
   .option("checkpointLocation",
"abfss://healthcare@vaishnavidata.dfs.core.windows.net/chkpt/bronze_labsd
ata")
   .outputMode("append")
    .trigger(availableNow=True)
   .table(f"{catalog}.{schema}.bronze_labsdata")
```

```
)
```

2) Bronze Patients Data

```
bronze_patientsdata = (spark.readStream
    .format("cloudFiles")
    .option("cloudFiles.format", "csv")
    .option("header","true")
    .option("cloudFiles.schemaLocation",
"abfss://healthcare@vaishnavidata.dfs.core.windows.net/schema/bronze_patientsdata_new")
    .option("cloudFiles.inferColumnTypes", "true")
    .load(raw_patients_data)
    .withColumn("ingest_file", col("_metadata.file_path"))
)
(bronze_patientsdata.writeStream
   .format("delta")
   .option("checkpointLocation",
"abfss://healthcare@vaishnavidata.dfs.core.windows.net/chkpt/bronze_patientsdata_new")
   .outputMode("append")
   .trigger(availableNow=True)
   .table(f"{catalog}.{schema}.bronze_patientsdata")
)
```

3) Bronze Visits Data

```
bronze_visitsdata = (spark.readStream
    .format("cloudFiles")
    .option("cloudFiles.format", "csv")
    .option("header","true")
    .option("cloudFiles.schemaLocation",
"abfss://healthcare@vaishnavidata.dfs.core.windows.net/schema/bronze_visitsdata")
    .option("cloudFiles.inferColumnTypes", "true")
    .option("cloudFiles.schemaHints","admission_date DATE, discharge_date DATE")
```

```
    .load(raw_visits_data)
    .withColumn("ingest_file", col("_metadata.file_path"))
)

(bronze_visitsdata.writeStream
   .format("delta")
   .option("checkpointLocation",
"abfss://healthcare@vaishnavidata.dfs.core.windows.net/chkpt/bronze_visitsdata"
)
   .outputMode("append")
   .trigger(availableNow=True)
   .table(f"{catalog}.{schema}.bronze_visitsdata")
)
```

4) Bronze Vitals Data

```
bronze_vitalsdata = (spark.readStream
    .format("cloudFiles")
    .option("cloudFiles.format", "csv")
    .option("header","true")
    .option("cloudFiles.schemaLocation",
"abfss://healthcare@vaishnavidata.dfs.core.windows.net/schema/bronze_vitalsdata
")
    .option("cloudFiles.inferColumnTypes", "true")
    .option("cloudFiles.schemaHints","timestamp TIMESTAMP")
    .load(raw_vitals_data)
    .withColumn("ingest_file", col("_metadata.file_path"))
)

(bronze_vitalsdata.writeStream
   .format("delta")
   .option("checkpointLocation",
"abfss://healthcare@vaishnavidata.dfs.core.windows.net/chkpt/bronze_vitalsdata"
)
   .outputMode("append")
   .trigger(availableNow=True)
   .table(f"{catalog}.{schema}.bronze_vitalsdata")
)
```

5) Bronze Patient Doctor Data

```
bronze_patient_doctor = (spark.readStream
    .format("cloudFiles")
    .option("cloudFiles.format", "csv")
    .option("header","true")
    .option("cloudFiles.schemaLocation",
"abfss://healthcare@vaishnavidata.dfs.core.windows.net/schema/bronze_patient_doctor")
    .option("cloudFiles.inferColumnTypes", "true")
    # .option("cloudFiles.schemaHints","timestamp TIMESTAMP")
    .load(raw_patient_doctor)
    .withColumn("ingest_file", col("_metadata.file_path"))
)

(bronze_patient_doctor.writeStream
   .format("delta")
   .option("checkpointLocation",
"abfss://healthcare@vaishnavidata.dfs.core.windows.net/chkpt/bronze_patient_doctor")
   .outputMode("append")
   .trigger(availableNow=True)
   .table(f"{catalog}.{schema}.bronze_patient_doctor")
)
```

Silver Tables

1) **Silver Patients Data**

```
patients_df =
spark.table("healthcare_vaishnavi.bronze.bronze_patientsdata")
# Replace 'INVALID_*' entries with None
patients_df_cleaned = patients_df.select([
   when(col(c).rlike("INVALID_.*"), "Unknown").otherwise(col(c)).alias(c)
   for c in patients_df.columns
])
```

```
# Cast age to int and filter out unrealistic values
patients_df_cleaned = patients_df_cleaned.withColumn("age",
col("age").cast("int"))
patients_df_cleaned = patients_df_cleaned.filter((col("age").isNotNull())
& (col("age") <= 120))

# Standardize gender values
patients_df_cleaned = patients_df_cleaned.withColumn("gender",
   when(col("gender").isin("M", "F", "Other"),
col("gender")).otherwise("Other")
)

# Fill remaining nulls
patients_df_cleaned = patients_df_cleaned.fillna({
   "age": 50,
   "gender": "Other",
   "diagnosis": "Undiagnosed",
   "prescription": "None"
})

# Drop rows with missing patient_id or name
patients_df_cleaned = patients_df_cleaned.dropna(subset=["patient_id",
"name"])
# Remove duplicates
patients_df_cleaned = patients_df_cleaned.dropDuplicates()
patients_df_cleaned = patients_df_cleaned.filter(col("prescription") !=
"Unknown")
# Ensure all columns match the target schema
patients_df_cleaned = patients_df_cleaned.select(
   col("patient_id").cast("string"),
   col("name").cast("string"),
   col("age").cast("int"),
   col("gender").cast("string"),
   col("diagnosis").cast("string"),
   col("prescription").cast("string")
)
```

```
patients_df_cleaned.write \
    .format("delta") \
    .mode("overwrite") \
    .option("overwriteSchema", "true") \
    .saveAsTable("healthcare_vaishnavi.silver.silver_patientsdata")
```

2) Silver Labs Data

```
labs_df = spark.table("healthcare_vaishnavi.bronze.bronze_labsdata")


labs_cleaned = labs_df.select([
    when(col(c).rlike("^INVALID_.*"),
"Unknown").otherwise(col(c)).alias(c)
    for c in labs_df.columns
])


labs_cleaned = labs_cleaned.dropna(subset=["patient_id",
"test_name","date"])


labs_cleaned = labs_cleaned.fillna({"unit": "Unspecified"})


labs_cleaned = labs_cleaned.withColumn("result",
col("result").cast("float"))
labs_cleaned = labs_cleaned.filter((col("result").isNotNull()) &
(col("result") >= 0) & (col("result") <= 1000))
labs_cleaned = labs_cleaned.dropDuplicates()

labs_cleaned = labs_cleaned.select(
    col("patient_id").cast("string"),
    col("test_name").cast("string"),
    col("result").cast("float"),
    col("unit").cast("string"),
    col("date").cast("date")
)
```

```
labs_cleaned.write \
   .mode("overwrite") \
   .format("delta") \
   .saveAsTable("healthcare_vaishnavi.silver.silver_labsdata")
```

3) Silver visits data

```
visits_df = spark.table("healthcare_vaishnavi.bronze.bronze_visitsdata")

visits_cleaned = visits_df.select([
   when(col(c).rlike("^INVALID_.*"),
"Unknown").otherwise(col(c)).alias(c)
   for c in visits_df.columns
])


visits_cleaned = visits_cleaned.dropna(subset=["patient_id",
"admission_date"])


visits_cleaned = visits_cleaned.fillna({"reason": "Unspecified"})
visits_cleaned = visits_cleaned.filter(col("reason") != "Unknown")


visits_cleaned = visits_cleaned.dropDuplicates()

visits_cleaned = visits_cleaned.select(
   col("patient_id").cast("string"),
   col("admission_date").cast("date"),
   col("discharge_date").cast("date"),
   col("reason").cast("string")
)
visits_cleaned = visits_cleaned.withColumn(
    "is_active",
    when(col("discharge_date").isNull(), True).otherwise(False)
)
visits_cleaned.write \
```

```
    .mode("overwrite") \
    .format("delta") \
    .saveAsTable("healthcare_vaishnavi.silver.silver_visitsdata")
```

4) Silver vitals data

```
df = spark.table("healthcare_vaishnavi.bronze.bronze_vitalsdata ")


df_cleaned = df.select([
    when(col(c).rlike("^INVALID_.*"),
"Unknown").otherwise(col(c)).alias(c)
    for c in df.columns
])


df_cleaned = df_cleaned.dropna(subset=["patient_id", "timestamp"])


df_cleaned = df_cleaned.fillna({
    "device_id": "Unspecified",
    "vital_type": "Unspecified"
})


# Cast vital_value to float and filter out unrealistic values
df_cleaned = df_cleaned.withColumn("vital_value",
col("vital_value").cast("float"))
df_cleaned = df_cleaned.filter((col("vital_value").isNotNull()) &
(col("vital_value") >= 0) & (col("vital_value") <= 1000))


df_cleaned = df_cleaned.filter(col("vital_type") != "Unspecified")
df_cleaned = df_cleaned.filter(col("device_id") != "Unspecified")


df_cleaned = df_cleaned.dropDuplicates()
```

```
# Ensure schema consistency
df_cleaned = df_cleaned.select(
    col("patient_id").cast("string"),
    col("device_id").cast("string"),
    col("timestamp").cast("timestamp"),
    col("vital_type").cast("string"),
    col("vital_value").cast("float")
)


df_cleaned.write \
    .format("delta") \
    .mode("overwrite") \
    .option("overwriteSchema", "true") \
    .saveAsTable("healthcare_vaishnavi.silver.silver_vitalsdata")
```

5) Silver patient doctor map data

```
doctor_map_df =
spark.table("healthcare_vaishnavi.bronze.bronze_patient_doctor")

doctor_map_cleaned = doctor_map_df.select([
   when(col(c).rlike("INVALID_.*"), "Unknown").otherwise(col(c)).alias(c)
   for c in doctor_map_df.columns
])

doctor_map_cleaned =
doctor_map_cleaned.dropna(subset=["patient_id","doctor_id"])

doctor_map_cleaned = doctor_map_cleaned.fillna({"care_team":
"Unassigned"})
doctor_map_cleaned = doctor_map_cleaned.filter(col("care_team") !=
"Unknown")
doctor_map_cleaned = doctor_map_cleaned.filter(col("doctor_id") !=
"Unknown")

doctor_map_cleaned = doctor_map_cleaned.dropDuplicates()
```

```
doctor_map_cleaned = doctor_map_cleaned.select(
    col("patient_id").cast("string"),
    col("doctor_id").cast("string"),
    col("care_team").cast("string")
)


doctor_map_cleaned.write \
    .format("delta") \
    .mode("overwrite") \
    .saveAsTable("healthcare_vaishnavi.silver.silver_patient_doctor")
```

Gold Tables

1) Patient Profile Data in Raw and clean(for BI)

```
# Base join
patient_profile_base = (
    spark.table("healthcare_vaishnavi.silver.silver_patientsdata")

.join(spark.table("healthcare_vaishnavi.silver.silver_patient_doctor"),
"patient_id", "left")
)

# Cleaned version for dashboards
patient_profile_clean = (
    patient_profile_base
    .fillna({"doctor_id": "Unassigned", "care_team": "General"})
)

# Save as Gold table for BI
patient_profile_clean.write.mode("overwrite").saveAsTable(
    "healthcare_vaishnavi.gold.gold_patient_profile_clean"
)

# Raw version (keep nulls for ML/analytics)
patient_profile_raw = patient_profile_base
```

```
# Save as Gold table for ML
patient_profile_raw.write.mode("overwrite").saveAsTable(
    "healthcare_vaishnavi.gold.gold_patient_profile_raw"
)
```

2) Latest Lab Results

```
from pyspark.sql import Window


# Window to pick latest lab test per patient
w = Window.partitionBy("patient_id",
"test_name").orderBy(col("date").desc())

lab_trends_df = (
    spark.table("healthcare_vaishnavi.silver.silver_labsdata")
    .withColumn("rn", row_number().over(w))
    .filter(col("rn") == 1)
    .drop("rn")
)

# Save as Gold table
lab_trends_df.write.format("delta").mode("overwrite").saveAsTable("health
care_vaishnavi.gold.latest_lab_results")
```

3) Visits Summary

```
visit_summary_df = (
    spark.table("healthcare_vaishnavi.silver.silver_visitsdata")
    .groupBy("patient_id")
    .agg(
        count("*").alias("total_visits"),
        sum(when(col("is_active"),
1).otherwise(0)).alias("active_visits"),
        sum(when(~col("is_active"),
1).otherwise(0)).alias("completed_visits"),
        avg(datediff("discharge_date",
"admission_date")).alias("avg_stay_length"),
```

```
        max("discharge_date").alias("last_discharge")
    )
)


visit_summary_df.write.format("delta").mode("overwrite").saveAsTable("hea
lthcare_vaishnavi.gold.visit_summary")
```

4) Vitals summary

```
vitals_daily_df = (
    spark.table("healthcare_vaishnavi.silver.silver_vitalsdata")
    .withColumn("date", to_date("timestamp"))
    .groupBy("patient_id", "vital_type", "date")
    .agg(
         avg("vital_value").alias("avg_value"),
         max("vital_value").alias("max_value"),
         min("vital_value").alias("min_value")
    )
)

vitals_daily_df.write.format("delta").mode("overwrite").saveAsTable("hea
lthcare_vaishnavi.gold.vitals_daily")
```

5) Risk Features

```
from pyspark.sql import functions as F

# Rename 'date' in vitals
vitals_df = spark.table("healthcare_vaishnavi.gold.vitals_daily") \
                 .withColumnRenamed("date", "vitals_date")

# Rename 'date' in labs
labs_df = spark.table("healthcare_vaishnavi.gold.latest_lab_results") \
               .withColumnRenamed("date", "lab_date")

risk_features_df = (
    spark.table("healthcare_vaishnavi.gold.gold_patient_profile_raw")
    .join(vitals_df, "patient_id", "left")
```

```
    .join(labs_df, "patient_id", "left")
    .join(spark.table("healthcare_vaishnavi.gold.visit_summary"),
"patient_id", "left")
)

risk_features_df.write.format("delta").mode("overwrite").saveAsTable(
    "healthcare_vaishnavi.gold.risk_features"
)
```

6) Medication Summary

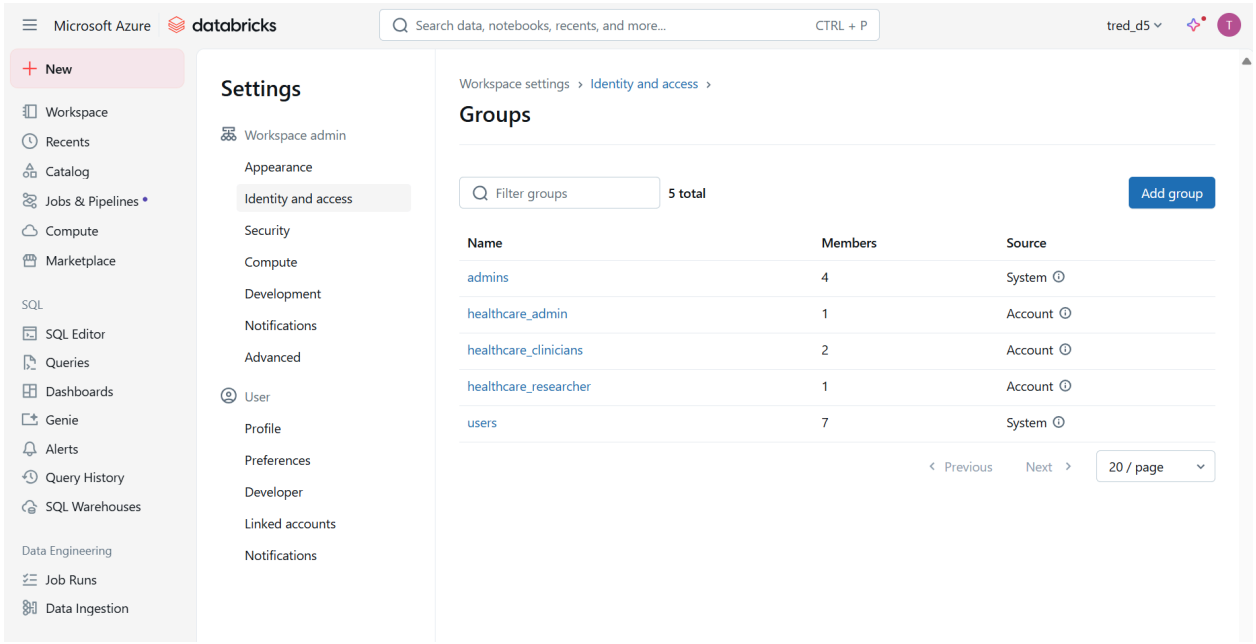
```
from pyspark.sql import functions as F

# Read silver patients table
patients_df =
spark.table("healthcare_vaishnavi.silver.silver_patientsdata")

# Split prescriptions into array (assuming comma-separated)
med_summary_df = (
    patients_df
    .withColumn("med_list", F.split(F.col("prescription"), ",\\s*"))
    .withColumn("diagnosis_list", F.split(F.col("diagnosis"), ",\\s*"))
    .withColumn("total_medications", F.size(F.col("med_list")))
    .withColumn("latest_prescription", F.element_at(F.col("med_list"),
-1))  # last item in list
    .select("patient_id", "diagnosis_list", "total_medications",
"latest_prescription")
)

# Save as Gold table
med_summary_df.write.format("delta").mode("overwrite").saveAsTable("healt
hcare_vaishnavi.gold.medication_summary")
```

Permissions
Created groups in databricks workspace.



Grant all permissions to admin

```
%sql
GRANT ALL PRIVILEGES ON SCHEMA healthcare_vaishnavi.bronze TO
`healthcare_admin`;
GRANT ALL PRIVILEGES ON SCHEMA healthcare_vaishnavi.silver TO
`healthcare_admin`;
GRANT ALL PRIVILEGES ON SCHEMA healthcare_vaishnavi.gold   TO
`healthcare_admin`;
```

If we had login mails then we could have allowed particular doctor to view only his patient.

```
%sql
CREATE OR REPLACE TABLE healthcare_vaishnavi.gold.doctor_user_mapping (
    doctor_id STRING,          -- used in patient records
    user_principal_name STRING -- login/email that current_user() returns
);

-- Insert 5 doctors
INSERT INTO healthcare_vaishnavi.gold.doctor_user_mapping VALUES
  ('D001', 'vaishnavi.m@hospital.org'),
  ('D002', 'amit.k@hospital.org'),
  ('D003', 'sneha.r@hospital.org'),
```

```
  ('D004', 'rahul.s@hospital.org'),
  ('D005', 'priya.t@hospital.org');
```

```
%sql
CREATE OR REPLACE VIEW healthcare_vaishnavi.gold.secure_patient_profile AS
SELECT p.*
FROM healthcare_vaishnavi.gold.gold_patient_profile_clean p
JOIN healthcare_vaishnavi.gold.doctor_user_mapping m
  ON p.doctor_id = m.doctor_id
WHERE m.user_principal_name = current_user();
```

Grant only limited permission to clinicians

```
%sql
GRANT SELECT ON VIEW healthcare_vaishnavi.gold.secure_patient_profile TO
`healthcare_clinicians`;
```

Hiding PII from Researchers

```
%sql
CREATE OR REPLACE VIEW healthcare_vaishnavi.gold.research_patient_profile AS
SELECT patient_id,
       SHA2(name, 256) AS anonymized_name,
       age,
       gender,
       diagnosis,
       prescription
FROM healthcare_vaishnavi.gold.gold_patient_profile_clean;
```

```
%sql
GRANT SELECT ON VIEW healthcare_vaishnavi.gold.research_patient_profile TO
`healthcare_researcher`;
```

Only admin can view all the information not everyone.

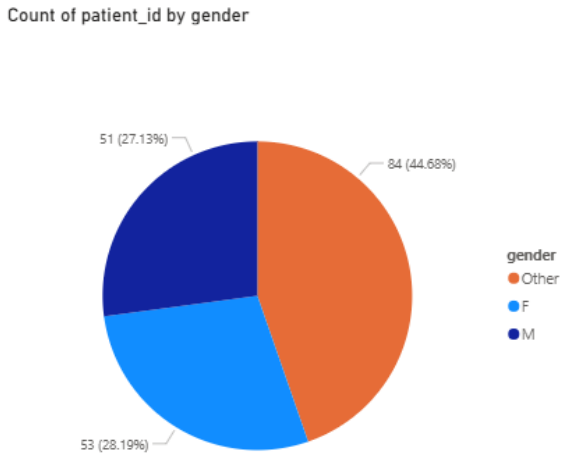
```sql
%sql
CREATE OR REPLACE FUNCTION healthcare_vaishnavi.gold.mask_name(name STRING)
RETURNS STRING
RETURN CASE
  WHEN is_member('healthcare_admin') THEN name
  ELSE concat('Patient_', sha2(name, 256))
END;
```

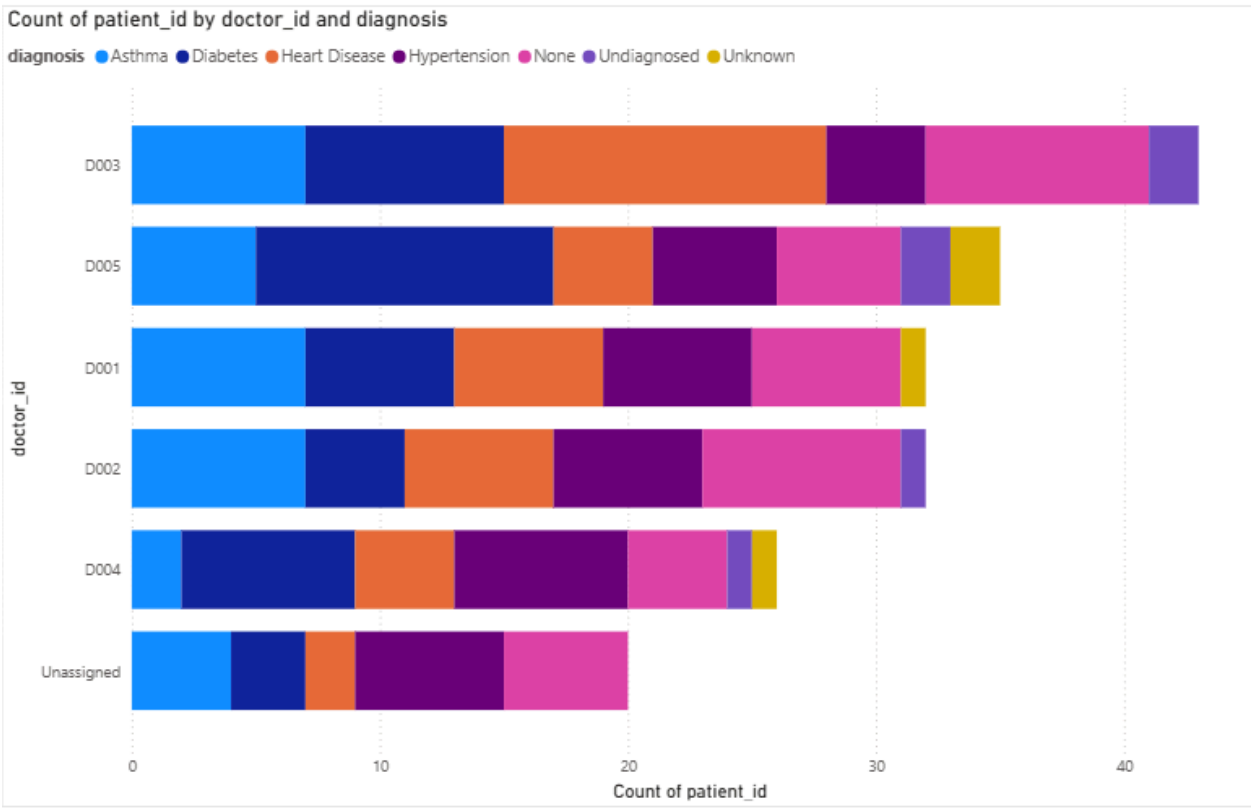
```sql
%sql
SELECT patient_id, healthcare_vaishnavi.gold.mask_name(name), age, gender
FROM healthcare_vaishnavi.silver.silver_patientsdata;
```

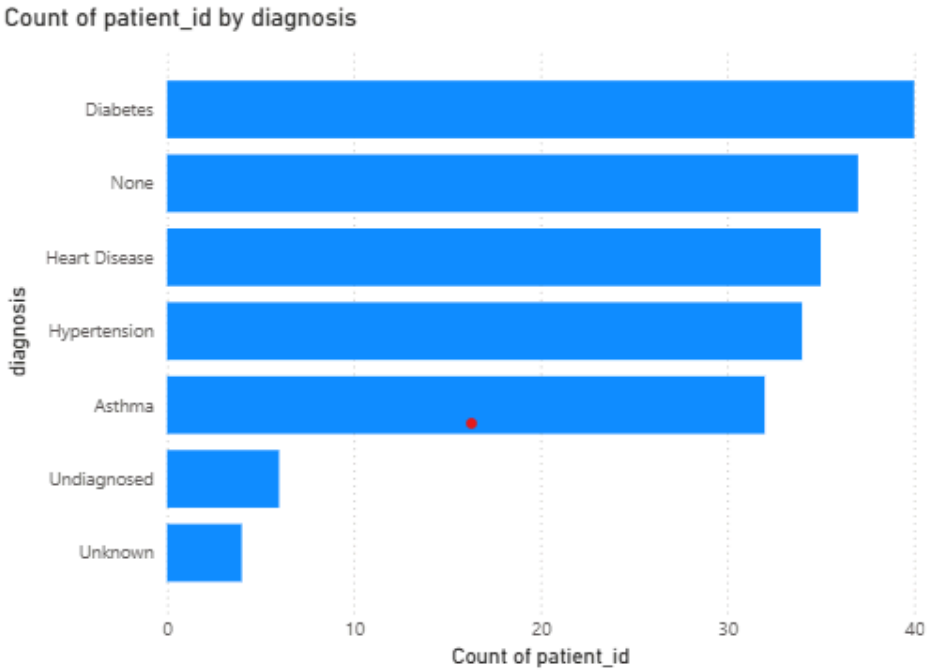
Retention Policy of 10 years

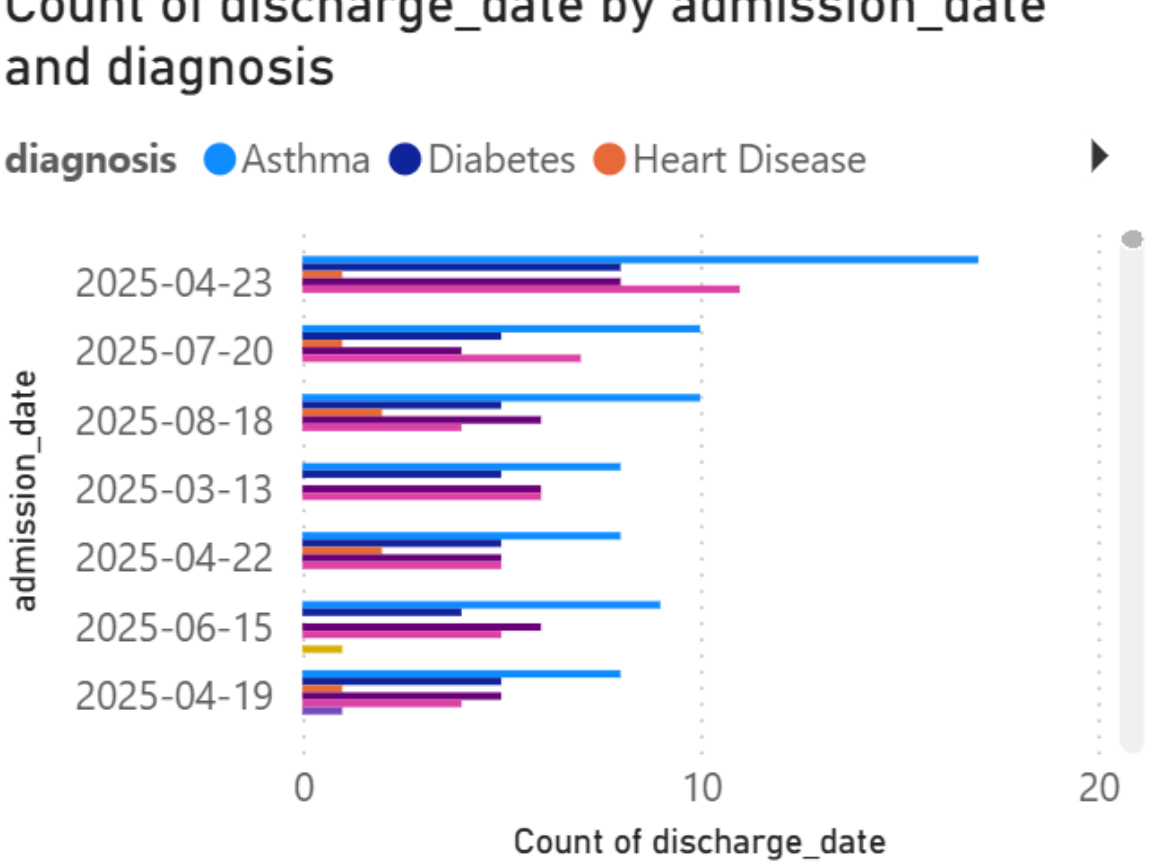
```sql
%sql
ALTER TABLE healthcare_vaishnavi.silver.silver_patientsdata
SET TBLPROPERTIES (
  'delta.logRetentionDuration' = '3650 days',
  'delta.deletedFileRetentionDuration' = '3650 days'
);
```

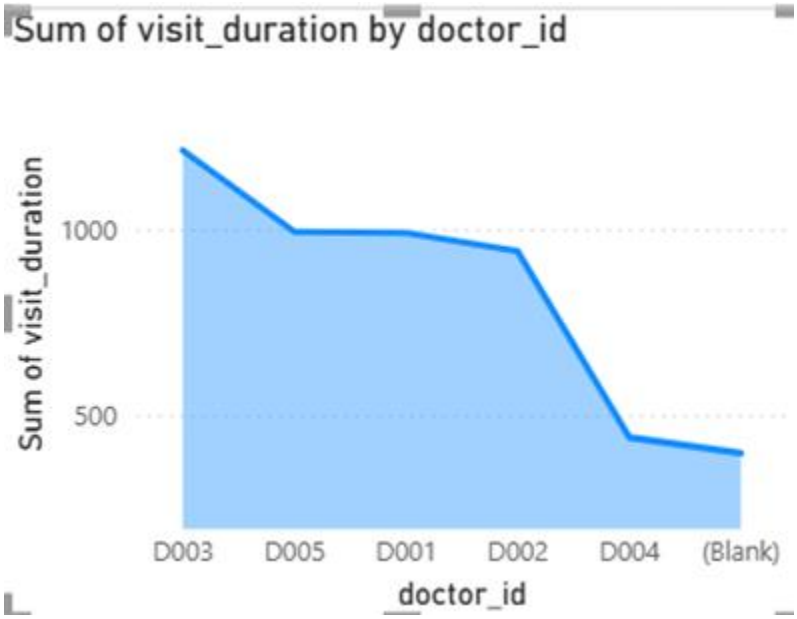
PowerBi visualizations

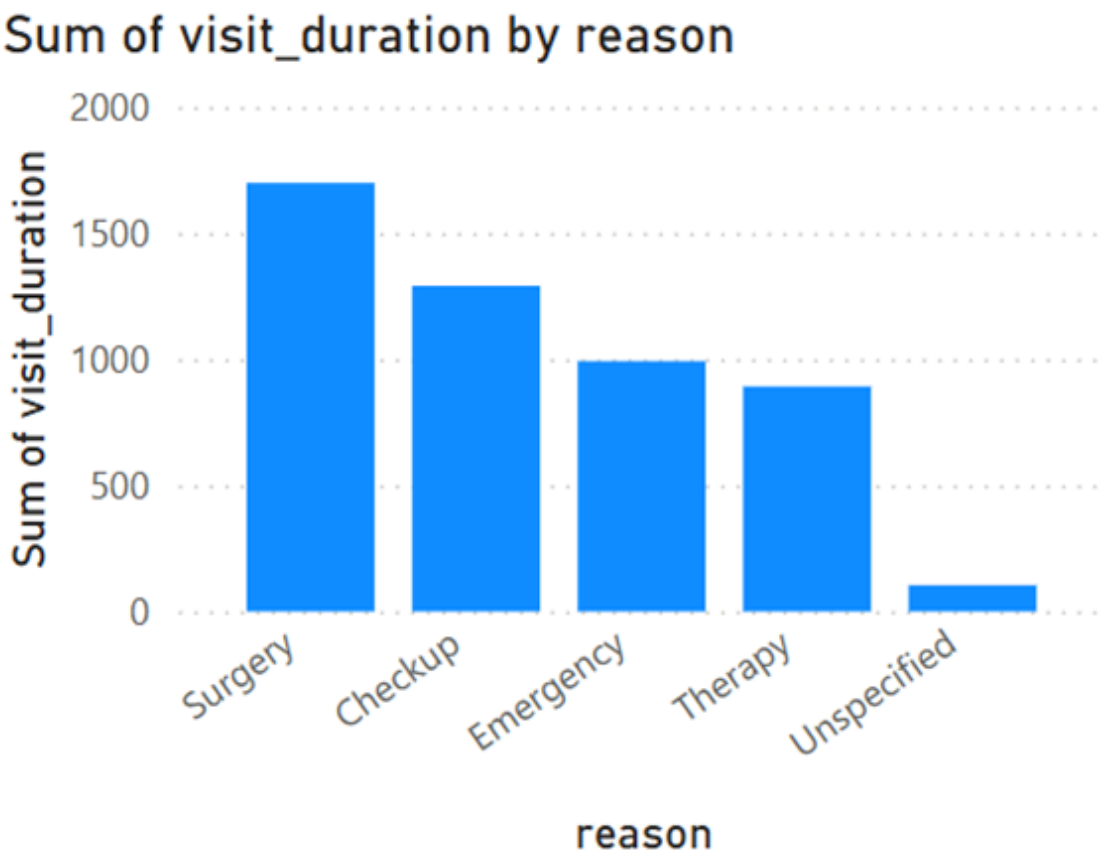

Count of patient_id by gender
51 (27.13%)
84 (44.68%)
53 (28.19%)
gender
Other
F
M

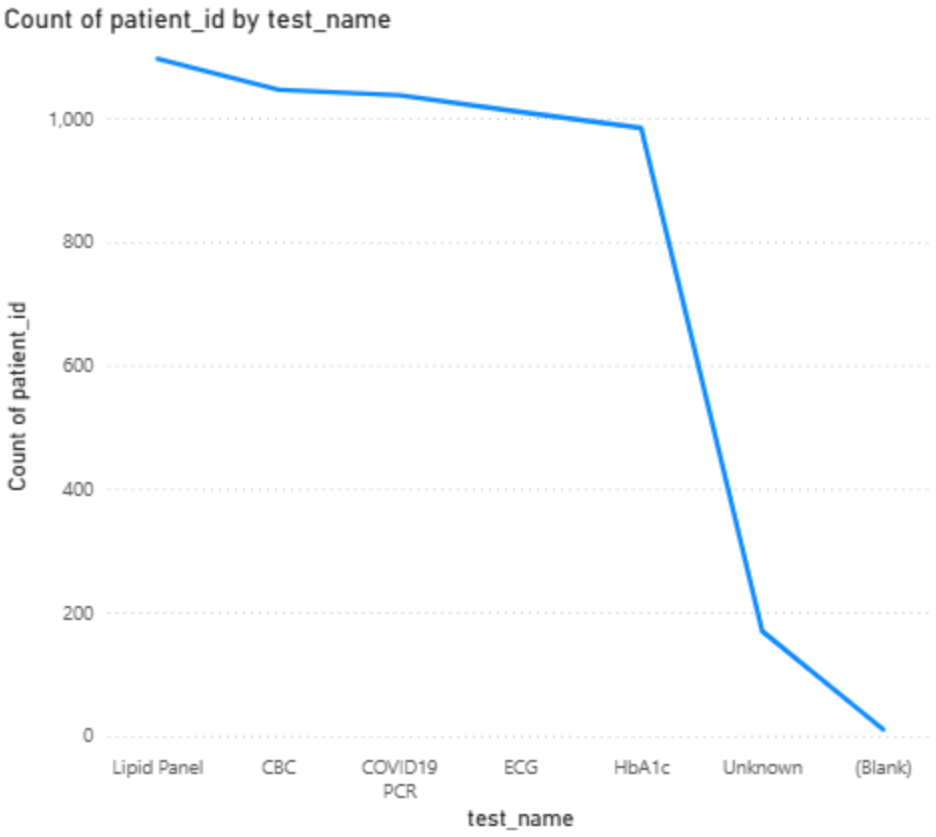

Count of patient_id by doctor_id and diagnosis
diagnosis
Asthma
Diabetes
Heart Disease
Hypertension
None
Undiagnosed
Unknown
doctor_id
D003
D005
D001
D002
D004
Unassigned
0
10
20
30
40
Count of patient_id

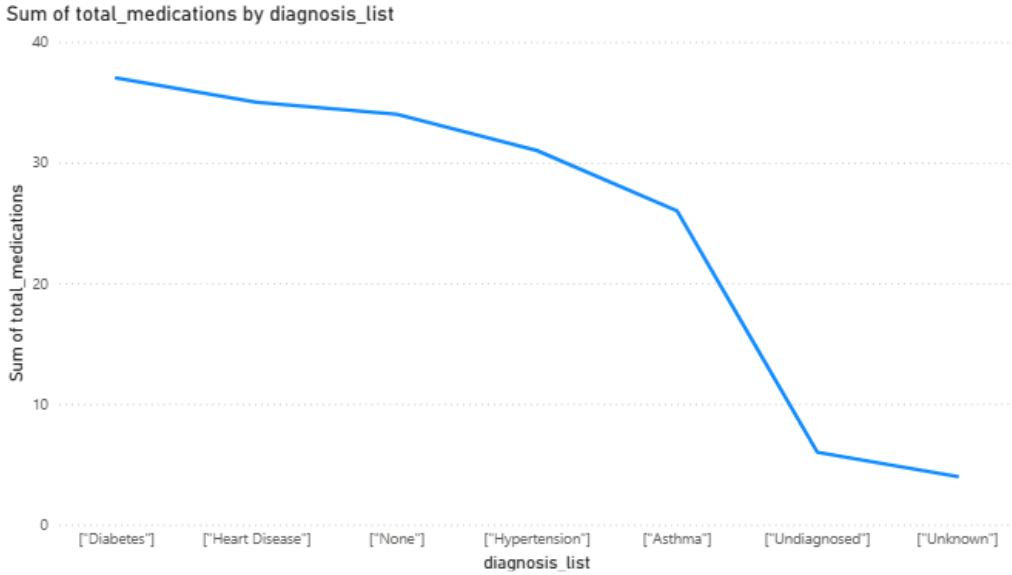

Count of patient_id by diagnosis
Diabetes
None
Heart Disease
Hypertension
Asthma
Undiagnosed
Unknown
diagnosis
0
10
20
30
40
Count of patient_id

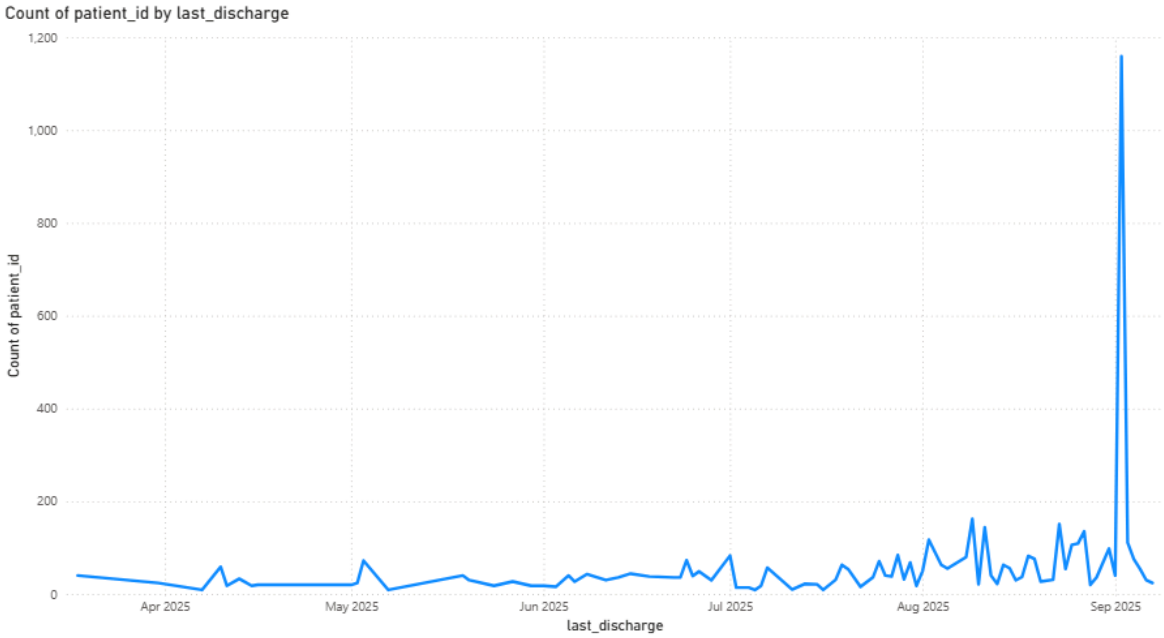

Count of discharge_date by admission_date and diagnosis
diagnosis
Asthma
Diabetes
Heart Disease
2025-04-23
2025-07-20
2025-08-18
2025-03-13
2025-04-22
2025-06-15
2025-04-19
admission_date
0
10
20
Count of discharge_date


Sum of visit_duration by doctor_id
Sum of visit_duration
1000
500
D003
D005
D001
D002
D004
(Blank)
_doctor_id


Sum of visit_duration by reason
Sum of visit_duration
2000
1500
1000
500
0
Surgery
Checkup
Emergency
Therapy
Unspecified
reason

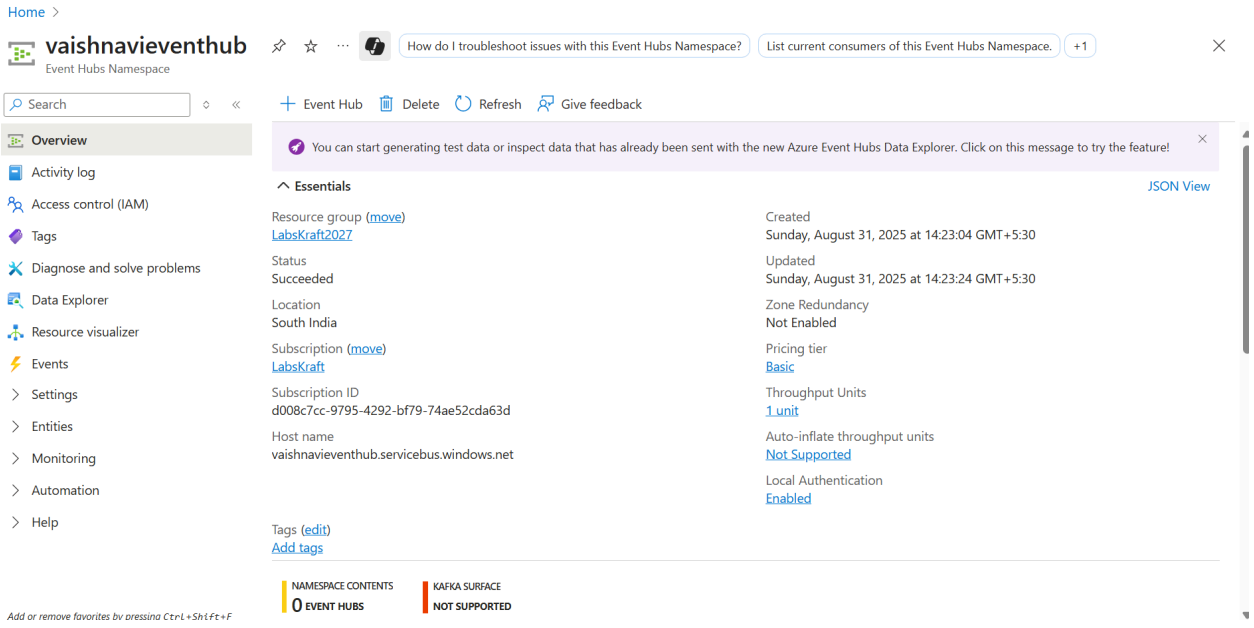

Count of patient_id by test_name
1,000
800
600
400
200
0
Count of patient_id
Lipid Panel
CBC
COVID19 PCR
ECG
HbA1c
Unknown
(Blank)
test_name

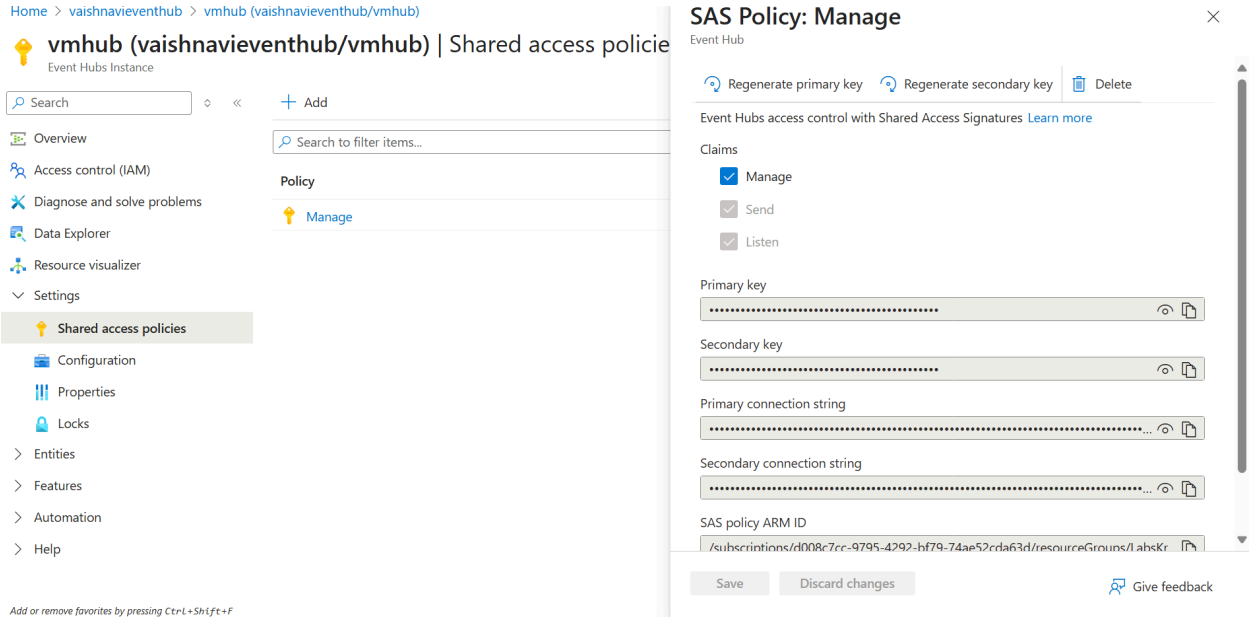

Sum of total_medications by diagnosis_list
40
30
20
10
0
Sum of total_medications
["Diabetes"]
["Heart Disease"]
["None"]
["Hypertension"]
["Asthma"]
["Undiagnosed"]
["Unknown"]
diagnosis_list


Count of patient_id by last_discharge
1,200
1,000
800
600
400
200
0
Count of patient_id
Apr 2025
May 2025
Jun 2025
Jul 2025
Aug 2025
Sep 2025
last_discharge

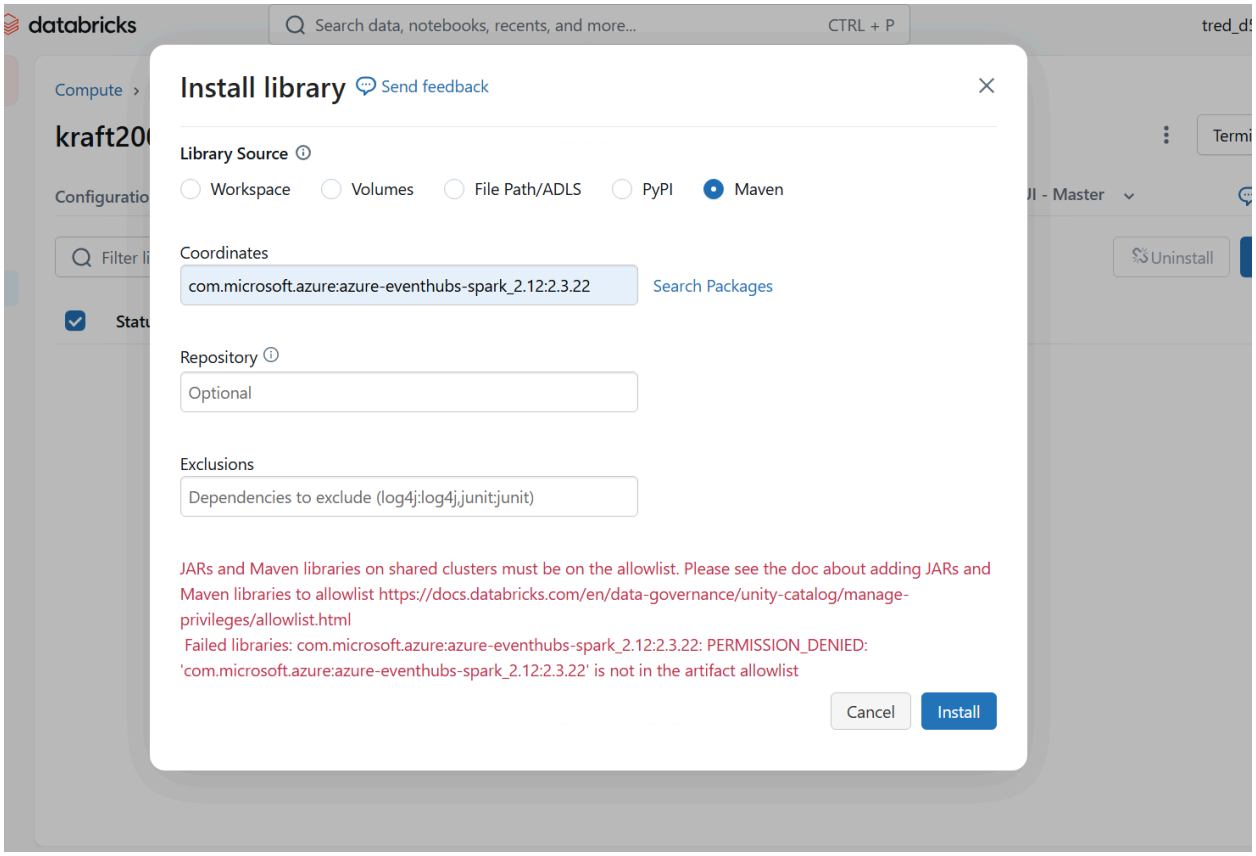
Streaming Vitals Data

Event Hub creation





In the connection -> add shared access policies ->copy the connection string.




```
# Databricks notebook (PySpark)
from pyspark.sql.types import StructType, StructField, StringType, DoubleType, TimestampType
from pyspark.sql.functions import from_json, col

# 1. declare schema for vitals JSON payload
vitals_schema = StructType([
    StructField("patient_id", StringType(), True),
    StructField("device_id", StringType(), True),
    StructField("vital_type", StringType(), True),      # e.g., heart_rate
    StructField("vital_value", DoubleType(), True),
    StructField("timestamp", TimestampType(), True),    # device timestamp
])

# 2. Event Hubs configuration
eventhub_conf = {

"Endpoint=sb://vaishnavieventhub.servicebus.windows.net/;SharedAccessKeyName=Ma
```

```
nage;SharedAccessKey=H/7hwApDmeZLfHHkYp4QxhQcrsnzmGnxt+AEhJEugow=;EntityPath=vm
hub"
}


# 3. Read stream
eh_raw = (spark.readStream
          .format("eventhubs")
          .option("eventhubs.connectionString",connection_string)
          .load()
         )

# 4. Extract body as string, parse JSON
raw_vitals = (eh_raw
              .selectExpr("cast(body as string) as body_str")
              .select(from_json(col("body_str"), vitals_schema).alias("data"))
              .select("data.*")
              .withColumn("ingest_received_ts", col("event_time"))  # optional
extra ts
             )

# 5. Write to Bronze Delta table with checkpointing
bronze_table = "healthcare_vaishnavi.bronze.bronze_vitals_eventhub"
checkpoint_path =
"abfss://healthcare@vaishnavidata.dfs.core.windows.net/chkpt/bronze_vitals_even
thub"

(bronze_raw := raw_vitals).writeStream \
    .format("delta") \
    .option("checkpointLocation", checkpoint_path) \
    .outputMode("append") \
    .trigger(processingTime="10 seconds") \    # micro-batches every 10s; tune
to get <30s alerts
    .table(bronze_table)
```

```
> [UNSUPPORTED_STREAMING_SOURCE_PERMISSION_ENFORCED] Data source eventhubs is not supported as a streaming source on
a shared cluster. SQLSTATE: 0A000
```